Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Multifamily Housing SaaS · Java · Python · Microservices · Event-Driven · AWS · AI Agents · LangGraph · FastAPI · PostgreSQL

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PROFESSIONAL SUMMARY

8 years building multifamily housing SaaS at Entrata — apartment leasing, maintenance requests, resident communications, pricing, financial settlement, and document workflows at 100K+ unit scale. EliseAI is putting AI on top of exactly these workflows. I've built them from the inside. Java + Python + microservices + event-driven architecture (RabbitMQ) + AWS = direct stack. Built Leasel: 8-agent LangGraph AI platform for property operations triage, RAG over lease documents, and automated HITL escalation. Built AgentProbe: open-source AI evaluation infrastructure. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and data pipelines; surfacing insights into application-facing APIs and reporting layers.

Magizhs Tech Solution May 2025 – Aug 2025

AI/ML Engineer Intern

- Built agentic automation system (LangGraph + FastAPI) for real estate stakeholders — shipped to real client, iterated weekly on customer feedback. Small team, fast pace, full ownership from day one.

Entrata (PropTech / Multifamily Housing SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Built apartment leasing, maintenance, and resident communication systems** used by 100K+ housing units — the exact workflows EliseAI automates: reservation management, maintenance request routing, tenant communication, document workflows, and financial settlement. Owned end-to-end from schema design through production monitoring and incident response. Zero compliance violations across 7 years.
- **Built event-driven automation system (RabbitMQ)** for processing account state changes across thousands of tenant records — retry logic, idempotency guarantees, owned through production incidents. Deployed 20+ modules across 120+ Agile sprints with no playbook and no hand-holding.
- **Owned multi-tenant PostgreSQL schemas** for leasing, financial settlement, and document workflows — 200+ SQL queries optimized (30% performance gain); schema design supporting data isolation across 100K+ units.
- Microservices redesign (Docker + Java + Python) — 50% scalability improvement, 40% deployment time reduction; mentored 5+ engineers; 80% manual operational workload eliminated through Python/Bash automation pipelines.

Entrata India, Pune Jul 2017 – Sep 2018

Associate Software Engineer

- Built housing and property management modules (Java, Python, PostgreSQL) across leasing, document management, and compliance workflows; led AWS S3 migration.

PROJECTS

Leasel: AI-Powered Property Operations Platform MULTIFAMILY AI LANGGRAPH

Python · LangGraph (8-agent) · FastAPI · OpenAI API · RAG · OCR | github.com/venkatalakshmiparimala/Lease_complaince_rag

- 8-agent AI system automating property operations triage — routes support tickets to specialized agents (configuration check, historical incident lookup, document analysis), produces evidence-backed root-cause conclusions, escalates to human when confidence is low. Same intelligent automation pattern as EliseAI's resident communication and maintenance triage agents. RAG over property documents; 93.6% precision, 87.9% F1 on 1,240 lease PDFs.

AgentProbe: AI Evaluation Infrastructure OPEN SOURCE CI-INTEGRATED

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI regression harness, model routing across OpenAI/Anthropic/Groq/Ollama. Built to catch AI quality regressions before they reach production users.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages & Frameworks: Java (Spring Boot), Python (8+ yrs), JavaScript, TypeScript, SQL, PHP, Bash — FastAPI, React.js, Django, REST APIs

Backend & Systems: Microservices, Event-Driven Architecture (RabbitMQ, Apache Kafka), Distributed Systems, Multi-Tenant Architecture, Idempotency, Fault-Tolerant Design, PostgreSQL, Redis, Celery, Temporal, Docker, Kubernetes

AI & Agents: LangChain, LangGraph, OpenAI API, Anthropic API, RAG Pipelines, Agentic Workflows, Tool Calling, HITL, LLM Evaluation, Embeddings, ChromaDB, MCP, Prompt Engineering

Cloud & Infra: AWS (S3, EC2, RDS, Lambda, Step Functions, Bedrock), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions), OpenTelemetry, Prometheus, Grafana, Observability, Performance Engineering